Laboratory 10

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EX01: PCA for Source Separation of Abdominal ECG Signals

Introduction

In this exercise we use PCA for the separation of maternal and fetal electrocardiography (ECG) signals in abdominal ECG (aECG) data recorded on the belly of a pregnant woman. Due to the low signal strengh of fetal ECG (fECG) signals it is an "algorithmic challenge" to properly separate fECG from much stronger maternal ECG (mECG) signals [1].

The present example uses a simplified version of the method proposed by Varanini et al. [2].

References

[1] R. Kahankova et al., “A Review of Signal Processing Techniques for Non-Invasive Fetal Electrocardiography,” IEEE Reviews in Biomedical Engineering, vol. 13, pp. 51–73, 2020, doi: 10.1109/RBME.2019.2938061.

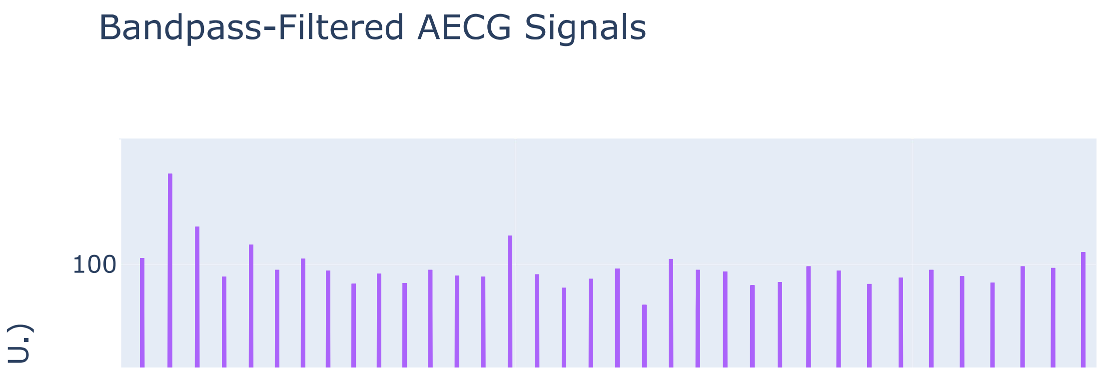
[2] M. Varanini, G. Tartarisco, L. Billeci, A. Macerata, G. Pioggia, and R. Balocchi, “An efficient unsupervised fetal QRS complex detection from abdominal maternal ECG,” Physiol. Meas., vol. 35, no. 8, pp. 1607–1619, Aug. 2014, doi: 10.1088/0967-3334/35/8/1607.

[3] Source of data: https://physionet.org/content/challenge-2013/1.0.0/

In [5]:
```
import numpy as np
import pandas as pd
from scipy.signal import filtfilt, butter
from sklearn.decomposition import PCA

import plotly.graph_objects as go
from plotly.offline import init_notebook_mode, iplot
from plotly.subplots import make_subplots
init_notebook_mode(connected=True)  # initiate notebook for offline plot

from mqrs_utils import cancel_mqrs
from ecgdetectors import Detectors
```

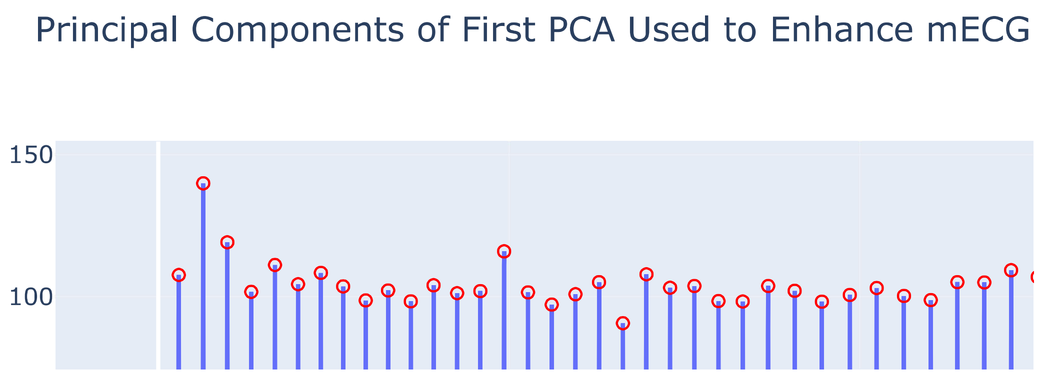
In [6]:
```python
# load abdominal ECG (aECG) data
# transformed from initial source: https://physionet.org/content/challenge-2013/
filename = 'aecg_a13.hdf5'
aecg = pd.read_hdf(filename, key='signals').values
fs = 1000
t = np.arange(aecg.shape[0]) / fs
```

In [7]:
```python
# bandpass filter data
b, a = butter(4, np.asarray([3, 45]), fs=fs, btype='bandpass')
aecg = filtfilt(b, a, aecg, axis=0)

# plot
fig = go.Figure()
for i in range(aecg.shape[1]):
    fig.add_trace(go.Scatter(x=t, y=aecg[:, i], name='AECG{:d}'.format(i)))
fig.update_xaxes(title='Time (s)')
fig.update_yaxes(title='AECG Amplitude (A.U.)')
fig.update_layout(title='Bandpass-Filtered AECG Signals')
fig.show()
```



In [8]:
```python
# apply first PCA for enhancing maternal ECG component
pca1 = PCA()
pc1 = pca1.fit_transform(aecg)

# maternal ECG as the first principal component, note that this
# remains a guess and would need to be automated in the final solution
maternal_ecg = pc1[:, 0]
```

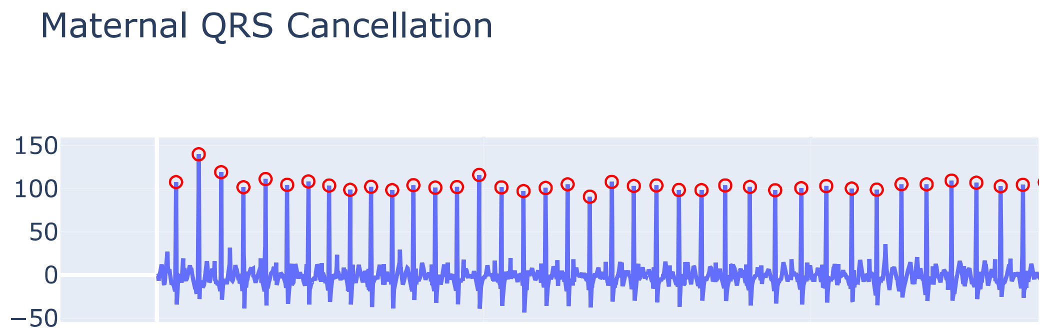
```
# detect maternal QRS peaks
mqrs_peaks = Detectors(fs).engzee_detector(maternal_ecg)

# plot
fig = go.Figure()
for i in range(pc1.shape[1]):
    fig.add_trace(go.Scatter(x=t, y=pc1[:, i], name='PC1[:,{:d}]'.format(i)))
    if i == 0:
        fig.add_trace(go.Scatter(x=t[mqrs_peaks], y=pc1[mqrs_peaks, i], name='mQ
                                 mode='markers', marker_color='red', marker_symb
fig.update_xaxes(title='Time (s)')
fig.update_yaxes(title='PC1 (A.U.)')
fig.update_layout(title='Principal Components of First PCA Used to Enhance mECG
fig.show()
```



In [9]:

```
# remove maternal QRS complexes from signal to obtain a best possible fetal ECG
x_residual, mecg_estimations = cancel_mqrs(fs, pc1, np.asarray(mqrs_peaks))

# plot
fig = make_subplots(rows=3, cols=1, shared_xaxes=True)
fig.add_trace(go.Scatter(x=t, y=pc1[:,0], name='Maternal ECG'), row=1, col=1)
fig.add_trace(go.Scatter(x=t[mqrs_peaks], y=pc1[mqrs_peaks, 0], name='mQRS-Peaks
                         legendgroup='mQRS', mode='markers', marker_symbol='circ
fig.add_trace(go.Scatter(x=t, y=mecg_estimations[:, 0], name='Interpolated mQRS
fig.add_trace(go.Scatter(x=t[mqrs_peaks], y=mecg_estimations[mqrs_peaks, 0], nam
                         legendgroup='mQRS', showlegend=False, mode='markers', m
fig.add_trace(go.Scatter(x=t, y=x_residual[:, 0], name='mQRS-free Signal'), row=
```

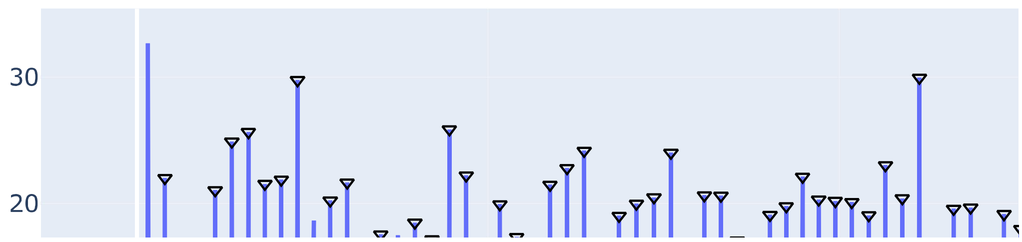
```
fig.add_trace(go.Scatter(x=t[mqrs_peaks], y=x_residual[mqrs_peaks, 0], name='mQR
                         legendgroup='mQRS', showlegend=False, mode='markers', m
fig.update_xaxes(title='Time (s)', row=3, col=1)
fig.update_layout(title='Maternal QRS Cancellation')
fig.show()
```



In [10]:

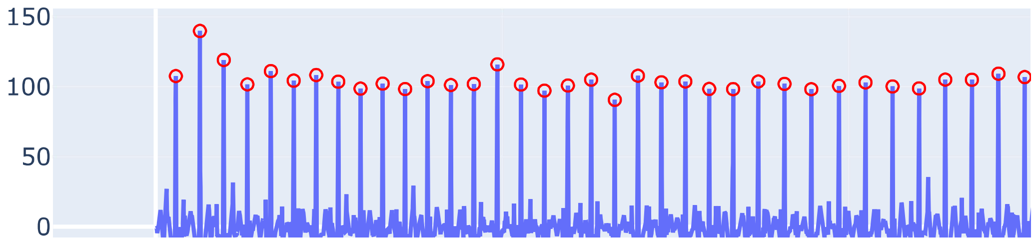
```
# apply second PCA for enhancing fetal ECG component in residual signal
pca2 = PCA()
pc2 = pca2.fit_transform(x_residual)

# fetal ECG as the first principal component, note that this
# remains a guess and would need to be automated in the final solution
fetal_ecg = pc2[:, 0]
# detect fetal QRS peaks
fqrs_peaks = Detectors(fs).engzee_detector(fetal_ecg)

# plot
fig = go.Figure()
for i in range(pc2.shape[1]):
    fig.add_trace(go.Scatter(x=t, y=pc2[:, i], name='PC2[:,{:d}]'.format(i)))
    if i == 0:
        fig.add_trace(go.Scatter(x=t[fqrs_peaks], y=pc2[fqrs_peaks, i], name='fQ
                                 mode='markers', marker_color='black', marker_sy
fig.update_xaxes(title='Time (s)')
fig.update_yaxes(title='PC2 (A.U.)')
fig.update_layout(title='Principal Components of Second PCA Used to Enhance fECG
fig.show()
```

Principal Components of Second PCA Used to Enhance fEC



In [11]:
```
# plot for summarizing all
fig = make_subplots(rows=2, cols=1, shared_xaxes=True)
# maternal ECG with mQRS
fig.add_trace(go.Scatter(x=t, y=maternal_ecg, name='Maternal ECG'), row=1, col=1
fig.add_trace(go.Scatter(x=t[mqrs_peaks], y=maternal_ecg[mqrs_peaks], name='mQRS
                         marker_color='red', mode='markers', marker_symbol='circ
# fetal ECG with fQRS
fig.add_trace(go.Scatter(x=t, y=fetal_ecg, name='Fetal ECG'), row=2, col=1)
fig.add_trace(go.Scatter(x=t[fqrs_peaks], y=fetal_ecg[fqrs_peaks], name='fQRS-Pe
                         marker_color='black', mode='markers', marker_symbol='tr
fig.update_xaxes(title='Time (s)', row=2, col=1)
fig.update_layout(title='Maternal vs. Fetal ECG')
fig.show()
```

Maternal vs. Fetal ECG



Exercise Questions

Please provide your answers directly below each question.

Question 1

Determine the maternal heart rate, both expressed in `Hz` and `beats/min`.

In [12]:
```
mqrs_time = np.array(mqrs_peaks)/fs
avg_mhr_hz = np.mean(1/np.diff(mqrs_time))
print('Average Maternal Heart Rate: {:.2f} Hz'.format(avg_mhr_hz))
print('Average Maternal Heart Rate: {:.2f} bpm'.format(avg_mhr_hz*60))

mqrs_time
```

```
Average Maternal Heart Rate: 1.37 Hz
Average Maternal Heart Rate: 82.08 bpm
```

Out[12]:
```
array([ 0.582,  1.277,  1.964,  2.646,  3.322,  3.982,  4.632,  5.266,
        5.91 ,  6.548,  7.195,  7.845,  8.515,  9.178,  9.855, 10.537,
       11.213, 11.89 , 12.562, 13.242, 13.912, 14.588, 15.279, 15.966,
       16.658, 17.384, 18.139, 18.912, 19.711, 20.476, 21.254, 22.019,
       22.776, 23.545, 24.31 , 25.067, 25.811, 26.49 , 27.162, 27.808,
       28.469, 29.145, 29.828, 30.533, 31.259, 32.041, 32.831, 33.594,
       34.37 , 35.143, 35.911, 36.725, 37.524, 38.346, 39.159, 39.983,
       40.802, 41.624, 42.427, 43.235, 44.058, 44.875, 45.673, 46.466,
       47.237, 48.01 , 48.78 , 49.545, 50.289, 51.061, 51.853, 52.623,
       53.413, 54.218, 55.001, 55.777, 56.551, 57.317, 58.038, 58.721,
       59.432])
```

Question 2

Determine the fetal heart rate, both expressed in `Hz` and `beats/min`.

In [13]:
```
fqrs_time = np.array(fqrs_peaks)/fs
avg_fhr_hz = np.mean(1/np.diff(fqrs_time))
print('Average Fetal Heart Rate: {:.2f} Hz'.format(avg_fhr_hz))
print('Average Fetal Heart Rate: {:.2f} bpm'.format(avg_fhr_hz*60))
```

```
Average Fetal Heart Rate: 1.91 Hz
Average Fetal Heart Rate: 114.86 bpm
```

Question 3

Determine the follwing three values:

- i) the average amplitude of the maternal QRS peaks (`mQRS`);
- ii) the average amplitude of the fetal QRS peaks (`fQRS`);
- iii) the ratio between the average amplitudes of i) `mQRS` and ii) `fQRS` peaks.

In [14]:
```
avg_mat_ecg_peaks = np.mean(maternal_ecg[mqrs_peaks])
avg_fetal_ecg_peaks = np.mean(fetal_ecg[fqrs_peaks])

print('Average Maternal ECG Peak Amplitude: {:.2f} A.U.'.format(avg_mat_ecg_peak
print('Average Fetal ECG Peak Amplitude: {:.2f} A.U.'.format(avg_fetal_ecg_peaks
print('Ratio Maternal/Fetal ECG Peak Amplitude: {:.2f}'.format(avg_mat_ecg_peaks
```

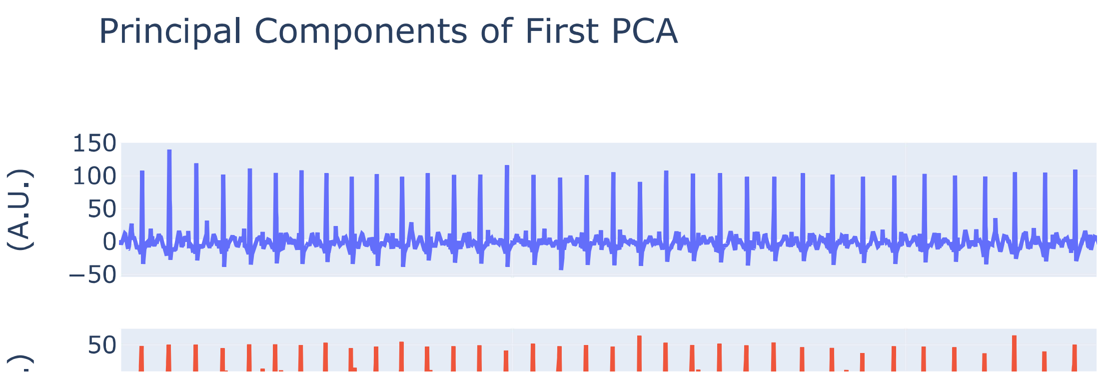
```
Average Maternal ECG Peak Amplitude: 103.25 A.U.
Average Fetal ECG Peak Amplitude: 20.50 A.U.
Ratio Maternal/Fetal ECG Peak Amplitude: 5.04
```

Question 4

How many of the principal components of the first PCA clearly show a maternal ECG signal? Which ones?

In [15]:
```
fig = make_subplots(rows=4, cols=1, shared_xaxes=True)
for i in range(pc1.shape[1]):
    fig.add_trace(go.Scatter(x=t, y=pc1[:, i], name='PC1[:,{:d}]'.format(i)), rc

fig.update_xaxes(title='Time (s)')
fig.update_yaxes(title='(A.U.)')
```

```
fig.update_layout(title='Principal Components of First PCA')
fig.show()
```



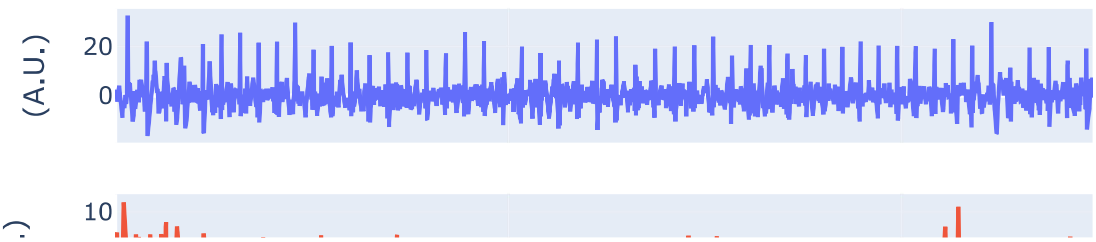
As shown in the previous plt, only the **first 2 principal components** of the 1st PCA clearly show the maternal ECG signal

Question 5

How many of the principal components of the second PCA clearly show a fetal ECG signal? Which ones?

In [16]:
```
fig = make_subplots(rows=4, cols=1, shared_xaxes=True)
for i in range(pc2.shape[1]):
    fig.add_trace(go.Scatter(x=t, y=pc2[:, i], name='PC2[:,{:d}]'.format(i)), rc

fig.update_xaxes(title='Time (s)')
fig.update_yaxes(title='(A.U.)')
fig.update_layout(title='Principal Components of Second PCA')
fig.show()
```

Principal Components of Second PCA



As shown in the previous plot, only the **first principal components** of the 2nd PCA clearly show the fetal ECG signal

Question 6

Not all of the fetal QRS peaks seem to be detected properly. Do you have an explanation why this happens and under which circumstances? Is it a problem of the fQRS detector, the mQRS cancellation or of another block of the algorithm?

By inspecting the code from `cancel_mqrs` it becomes clear that the main reason for inaccurate detection of fQRS peaks is related to the mQRS cancellation.

Since the mQRS cancellation works by simply subtracting the mQRS peak from the original signal, in situations in which there is a complete overlap between mQRS and fQRS then also the fQRS informration will be lost.

Hence, the resulting fECG signal will miss some peaks corresponding to the ones overlapping with the mother's ones.